Yan Cong

640 Oval Drive, West Lafayette, IN 47907
cong4@purdue.edu
https://yancong222.github.io/

Professional Positions

08/2023- **Assistant Professor**, *School of Languages and Cultures, Purdue University*
08/2023- **Affiliate**, *School of Interdisciplinary Studies, Purdue University*
01/2024- **Faculty Associate**, *Center on Aging and the Life Course, Purdue University*
2021-2023 **Postdoctoral Fellow**, *Feinstein Institutes, Northwell Health*
2015-2016 **Project Assistant**, *The Hong Kong Polytechnic University*

Education

07/2021 **Ph.D. in Linguistics**, *Michigan State University*
02/2015 **M.A. in Language Studies**, *The Hong Kong Baptist University*

Peer-Reviewed Journal Articles

15. **Cong, Yan**; Hu, Jingying; Reese, Timothy; Liu, Hui (2026). How robust are linguistic markers of aging? The case of aging-related social media text. *Natural Language Processing Journal*, 100203.
14. Cho, Sunghye; **Cong, Yan**; Mehta, Aarush; Nikzad, Amir H.; Berretta, Sarah A.; Behbehani, Leily M.; Liberman, Mark; Tang, Sunny X.. (2026). Unique signatures in verbal fluency task performance in schizophrenia and depression. *Schizophrenia Research: Cognition*, 43, 100407.
13. **Cong, Yan** and Lee, Jiyeon. (2025). Tracking priming-induced language recovery in aphasia with pre-trained language models. *Frontiers in Artificial Intelligence*, 8, 1668399.
12. Mehta, Aarush; Nikzad, Amir; **Cong, Yan**; Cho, Sunghye; Pradhan, Sameer; Tang, Sunny. X. (2025). Sentiment in speech is associated with symptom severity in psychosis. *Cognitive Neuropsychiatry*. 1–12.
11. **Cong, Yan**. (2025). Second language learning of degree expressions: A computational approach. *Cambridge Natural Language Processing*. 31(5), 1187–1209.
10. **Cong, Yan**. (2024). Manner implicatures in large language models. *Nature Scientific Reports*. 14, 29113.
9. Britton, James; **Cong, Yan**; Hsu, Yu-Yin; Chersoni, Emmanuele; and Blache, Philippe. (2024). On the Influence of Discourse Connectives on the Predictions of Humans and Language Models. *Frontiers in Human Neuroscience*. 18, 1363120.

8. **Cong, Yan**. (2024). Demystifying large language models in second language development research. *Computer Speech & Language*. 89, 101700.
7. **Cong, Yan**. (2024). AI Language Models: An Opportunity to Enhance Language Learning. *Informatics*. 11(3), 49.
6. **Cong, Yan**; LaCroix, Arianna N.; Lee, Jiyeon. (2024). Clinical efficacy of pre-trained large language models through the lens of aphasia. *Nature Scientific Reports*. 14, 15573.
5. Tang, Sunny X.; **Cong, Yan**; Mercep, Gwenyth; Bhatti, Mutahira; Serpe, Grace; Gromova, Valeria; Berretta, Sarah; John, Majnu; Liberman, Mark; Sinvani, Liron. (2023). Characterizing and Detecting Delirium with Clinical and Computational measures of Speech and Language Disturbance. *Journal of Psychiatry and Neuroscience*. 48(4), E255-E264.
4. Tang, Sunny X.; **Cong, Yan**; Nikzad, Amir; Mehta, Aarush; Cho, Sunghye; Hänsel, Katrin; Beretta, Sarah; Dhar, Aamina; Kane, John; Malhotra, Anil. (2022). Clinical and Computational Speech Measures are Associated with Social Cognition in Schizophrenia Spectrum Disorders. *Schizophrenia research*. 259, 28-37.
3. Nikzad, Amir; **Cong, Yan**; Beretta, Sarah; Hänsel, Katrin; Cho, Sunghye; Pradhan, Sameer; Behbehani, Leily; DeSouza, Danielle; Liberman, Mark; Tang, Sunny X. (2022). Who Does What to Whom? Graph Representations of Action-Predication in Speech Relate to Psychopathological Dimensions of Psychosis. *Nature Schizophrenia*. 8, 58.
2. Tang, Sunny X.; Hänsel, Katrin; **Cong, Yan**; Nikzad, Amir; Cho, Sunghye; Berretta, Sarah; Behbehani, Leily. (2022). Latent Factors of Speech and Language Disturbance and Relationships to Acoustic and Lexical Computational Features. *Schizophrenia Bulletin*. 49(Supplement_2), S93-S103.
1. **Cong, Yan**. (2022). Judge-dependence in Quality Nouns: A Semantic Analysis of the Mandarin Chinese "*you* NP" Structure. *Asian Language and Linguistics*. 2(2), 167-216.

Peer-Reviewed Conference Proceedings Articles

[M] Main ACL (Association for Computational Linguistics) venues
[W] Specialized workshops, indexed and included in the ACL Anthology
[R] Regional conferences, indexed and included in the ACL Anthology
[G] Graduate student author

13. **Cong, Yan**; Rayz, Julia. (2025). Language models demonstrate the good-enough processing seen in humans. In *Proceedings of the Annual Meeting of the Cognitive Science Society*. 47(0):4642-4649.
12. Li, Yue[G]; **Cong, Yan**; Francis, Elaine J. (2025). Beyond Binary Animacy: A Multi-Method Investigation of LMs' Sensitivity in English Object Relative Clauses. In *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics*. 184–196. ACL.

11. Hu, Jingying[G]; **Cong, Yan**. (2025). Modeling Chinese L2 Writing Development: The LLM-Surprisal Perspective. In *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics*. 172–183. ACL.

10[W]. **Cong, Yan**; Lee, Jiyeon; LaCroix, Arianna N. (2024). Leveraging pre-trained large language models for aphasia detection in English and Chinese speakers. In *Proceedings of the 6th Clinical Natural Language Processing Workshop (Clinical NLP)*. 238-245. ACL.

9[M]. A, Pranav; **Cong, Yan**; Chersoni, Emmanuele; Hsu, Yu-Yin.; Lenci, Alessandro. (2024). Comparing Static and Contextual Distributional Semantic Models on Intrinsic Tasks: An Evaluation on Mandarin Chinese Datasets. In *Proceedings of the Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*. 3610-3627. ACL.

8[W]. **Cong, Yan**; Chersoni, Emmanuele; Hsu, Yu-Yin; Blache, Philippe. (2023). Investigating the Effect of Discourse Connectives on Transformer Surprisal: Language Models Understand Connectives, Even So They Are Surprised. In *Proceedings of the 6th Conference on Blackbox NLP*. 222-232. ACL.

7[M]. **Cong, Yan**; Chersoni, Emmanuele; Hsu, Yu-Yin; Lenci, Alessandro. (2023). Are Language Models Sensitive to Semantic Attraction? A Study on Surprisal. In *Proceedings of the 12th Joint Conference on Lexical and Computational Semantics (*SEM)*. 141-148. ACL.

6[W]. **Cong, Yan**. (2022). Psycholinguistic Diagnosis of Language Models' Commonsense Reasoning. In *Proceedings of the First Workshop on Commonsense Representation and Reasoning (CSRR)*. 17–22. ACL.

5[W]. **Cong, Yan**. (2022). Pre-trained Language Models' Interpretation of Evaluativity Implicature: Evidence from Gradable Adjectives Usage in Context. In *Proceedings of the Second Workshop on Understanding Implicit and Underspecified Language*. 1–7. ACL.

4. **Cong, Yan**; Ngonyani, Deo. (2022). A syntactic analysis of the co-occurrence of stative and passive in Kiswahili. In Sibanda, Galen; Ngonyani, Deo; Choti, Jonathan; Biersteker, Ann (eds.). *Descriptive and theoretical approaches to African linguistics*. 165–181.

3[M]. Pandia, Lalchand; **Cong, Yan**; Ettinger, Allyson. (2021). Pragmatic competence of pre-trained language models through the lens of discourse connectives. In *Proceedings of the 25th Conference on Computational Natural Language Learning (CoNLL)*. 367–379. ACL.

2. **Cong, Yan**. (2019). The Non-canonical Argument Realization of Chinese Alternations. In *Proceedings of the 30th North American Conference on Chinese Linguistics (NACCL)*. 1, 82-99.

1[R]. **Cong, Yan**; Wee, Lian-Hee; Huang, Chu-Ren. (2015). The Invertible Construction in Chinese. In *Proceedings of the 29th Pacific Asia Conference on Language, Information and Computation (PACLIC)*. 206–211. ACL.

Peer-Reviewed Conference Abstracts

31. Li, Yue; **Cong, Yan**; Francis, Elaine J. (2026). *Illuminating Human Language Production via Mechanistic Interpretability of LM Circuits: Animacy Effects on Sentence Structure Choice.* In the 39th Annual Conference on Human Sentence Processing. March 26 - 28, 2026; MIT, Cambridge, MA.
30. Hu, Jingying; **Cong, Yan**; Francis, Elaine J. (2026). *Relative Clause Extraposition in English: An Information-theoretic Approach.* In the 39th Annual Conference on Human Sentence Processing. March 26 - 28, 2026; MIT, Cambridge, MA.
29. Fang, Shaohua; Li, Yue; **Cong, Yan**. (2026). *Can LLMs Read Between the Scopes? A Cross-Linguistic Study of Quantifier-Negation Interpretations.* In Linguistic Society of America (LSA) Annual Meeting. January 8 - 11, 2026; New Orleans, LA.
28. Fang, Shaohua; Li, Yue; **Cong, Yan**. *Every horse didn't jump: Scope preferences and ambiguity in LLM interpretation.* (2025). In Experimental Pragmatics Conference (XPRAG). September 17 - 19, 2025; University of Cambridge, UK.
27. **Cong, Yan** and Lee, Jiyeon. (2025). *Language models derived surprisal and structural priming in aphasia: Implications for language recovery.* In Academy of Aphasia Annual Meeting 2025. October 26 - 28, 2025; San Diego, CA.
26. Fang, Shaohua; Li, Yue; **Cong, Yan**. (2025). *Understanding Quantifier Scope with Large Language Models: How Many Children Climbed Trees?.* In Proceedings of the Annual Meeting of the Cognitive Science Society (Vol. 47). July 29 - August 1, 2025; San Francisco, CA.
25. **Cong, Yan**; LaCroix, Arianna N.; Jiyeon Lee. (2025). *Voice AI Symposium, Bridge2AI-Voice Consortium.* Enhancing Stroke-Induced Aphasia Assessment Using Interpretable linguistic biomarkers and Large Language Models. April 22 - 24, 2025; University of South Florida, CA.
24. Nikzad , Amir H.; **Cong, Yan**; Cho, Sunghye; Partlan, Ryan; Li, Jiefei; and Tang, Sunny X. (2025). *The 2025 Congress of the Schizophrenia International Research Society.* Evaluating Language Models and Preprocessing Strategies for Detecting Psychosis-Related Language Disturbances. March 29 - April 2, 2025; Chicago, IL.
23. Hu, Jingying and **Cong, Yan**. (2025). *The 8th International Forum on Linguistics and Chinese Education (IFOLCE) & The 32nd International Conference on Chinese Language Instruction.* Leveraging AI language models for Chinese L2 Writing Assessment. April 5 - 6, 2025; Princeton University, NJ.
22. Issa, Elsayed and **Cong, Yan**. (2025). *CALICO (Computer Assisted Language Instruction Consortium) Annual Conference.* Elicited Imitation Revisited: an open-sourced package for assessing L2 speech. May 27 - 31, 2025; San Diego State University, CA.
21. Fang, Shaohua; Li, Yue; **Cong, Yan**. (2025). *CLTA (The Chinese Language Teachers Association, USA, Inc.) Annual Conference.* Understanding Quantifier Scope with Large Language Models: How Many Children Climbed Trees? May 9 - 10, 2025; New York, NY.

20. Gautam, Sukrati; Yu, David J; **Cong, Yan**. (2024). *American Geophysical Union (AGU) Annual Meeting.* Enhancing Resilience in Reservoir Management: Addressing Logical Interdependencies through Framework Using Large Language Models and Natural Language Processing. December 9 - 13, 2024; Washington, D.C.
19. Issa, Elsayed; **Cong, Yan**; Scala, Carmela; Merschel, Lisa; Frattarola, Angela. Organizers: William, Jennifer; Fellin, Luciana. (2024). *Modern Language Association (MLA) Convention.* AI, NLP, and Computational Linguistics: Implications for Language Programs. January 4 - 7, 2024; Philadelphia, PA.
18. Nikzad, Amir H.; **Cong, Yan**; Berretta, Sarah; Behbehani, Leily; Pradhan, Sameer; Tang, Sunny X. (2023). *The 62nd Annual Meeting of the American College of Neuropsychopharmacology.* Disorganization Disrupts the Core: Localizing Clinical Dimensions of Speech and Language Disturbance Within Clause Structure Using Multi-Layered Semantic Graph Analysis. December 3 - 6, 2023; Tampa, FL.
17. Tang, Sunny X.; Nikzad, Amir H.; Spilka, MJ; **Cong, Yan**; Behbehani, Leily; Berretta, Sarah; Kane, JM; Malhotra, AK. (2023). *The 62nd Annual Meeting of the American College of Neuropsychopharmacology.* Linguistic risk score for schizophrenia: applying a polygenic risk approach to synthesize multivariate features from computational speech and language analysis. December 3 - 6, 2023; Tampa, FL.
16. **Cong, Yan**; Wolff, Phillip. (2022). *Annual Meeting of The Linguistic Society of America (LSA).* Inferring markedness from semantic weight: An approach using the T5 language model. Washington, D.C.
15. Nikzad, Amir; **Cong, Yan**; Hänsel, Katrin; Cho, Sunghye; Berretta, Sarah; Behbehani, Leily; Tang, Sunny X. (2022). *Annual Meeting of The Society of Biological Psychiatry (SOBP).* Modeling cross-diagnostic speech disturbance with semantic and structural speech graphs. Virtual presentation.
14. Hänsel, Katrin; Nikzad, Amir; **Cong, Yan**; Mehta, Aarush; Cho, Sunghye; Berretta, Sarah; Behbehani, Leily; Tang, Sunny X. (2022). *Annual Meeting of The Society of Biological Psychiatry (SOBP).* Predicting general psychopathology with multi-modal speech and language features. Virtual presentation.
13. **Cong, Yan** and Tang, Sunny X. (2022). *The 6th Annual Technology in Psychiatry Summit.* Leveraging state-of-the-art transformer language models to measure speech disturbance in schizophrenia. Harvard Medical School, Mclean Hospital.
12. **Cong, Yan**; Buccola, Brian. (2021). *The 45th annual Penn Linguistics Conference (PLC).* Alternatives and symmetry in non-scalar implicature: The case of Mandarin adjective constructions. Philadelphia, Pennsylvania.

11. Hänsel, Katrin; Cho, Sunghye; Berretta, Sarah; Mehta, Aarush; Nikzad, Amir; Guo, Jessica; **Cong, Yan**; Pradhan, Sameer; Fiumara, James; Liberman, Mark; Tang, Sunny X. (2021). *The 60th Annual meeting of American College of Neuropsychopharmacology (ACNP).* Associative dysregulation and impaired expressivity in schizophrenia: Latent structures of clinical and automated speech analysis features. Virtual presentation.
10. Smith, Kaylin; Ma, Ye; **Cong, Yan**; Beretta, Alan. (2019). *The Psycholinguistics in Iceland—Parsing and Prediction (PiPP).* Interactions between syntax and semantics: The semantic P600, University of Iceland. Reykjavík, Iceland.
9. Smith, Kaylin; Ma, Ye; **Cong, Yan**; Beretta, Alan. (2019). *The 32nd CUNY Conference on Human Sentence Processing.* Semantic attraction in sentence processing. University of Colorado Boulder. Boulder, Colorado.
8. **Cong, Yan**. (2019). *The Linguistic Society of America Linguistics Institute (LSA).* Information structure and the depth of language processing: a pilot study. University of California at Davis. California.
7. **Cong, Yan**. (2019). *International Workshop on Degrees and Grammar: An East Asian Perspective.* A subjective analysis of the Mandarin '*you*' structure. Nanjing University, China.
6. **Cong, Yan**. (2018). *The Annual Conference of the Chinese Language Teachers Association (CLTA).* The non-canonical argument realization of the Chinese alternations. Washington, D.C.
5. **Cong, Yan**. (2018). *The 12th Arizona Linguistics Circle (ALC).* Rethinking the implicit argument in Kiswahili. University of Arizona. Tucson, Arizona.
4. **Cong, Yan**. (2018). *The 2nd Crete Linguistics Summer School (CreteLing).* A preliminary semantic analysis of predicates of taste in Chinese. University of Crete. Greece.
3. Parrish, Alicia; Kelley, Patrick; Smith, Kaylin; **Cong, Yan**; Beretta, Alan. (2017). *Annual Meeting of The Society for the Neurobiology of Language (SNL).* Prosodic lengthening and boundary prediction in nominal compounds: An ERP study. Baltimore, Maryland.
2. **Cong, Yan**. (2017). *The Linguistic Society of America Linguistics Institute (LSA).* T1 and T0 Sandhi in Tongliao Mandarin: an OT approach. University of Kentucky. Lexington, Kentucky.
1. **Cong, Yan**. (2016). *The 28th North American Conference on Chinese Linguistics (NACCL).* Markedness theory and the acquisition of the Mandarin potential complement construction. Brigham Young University. Salt Lake City, Utah.

Invited Talks

2024 **Cong, Yan**. *Natural language processing and its implications in medicine and education.* Purdue System Thinkers (PurSysT) Research Colloquiums. Purdue University.

2024 **Cong, Yan**. *Language Understanding in Humans and AI.* Symposium: Current Ideas in Artificial Intelligence. The 50th Annual Meeting of the Society for Philosophy & Psychology. Purdue University.

2024 **Cong, Yan**. *Computational approach to linguistic biomarkers in aging.* Colloquium series. Center on Aging and the Life Course (CALC), Purdue University.

2023 **Cong, Yan**. *Large language models and their implications in text analysis.* Round table on *STEM in Asia.* Department of History, Purdue University.

Other Talks

2025 William, Jen; Issa, Elsayed; **Cong, Yan**. *Human side of AI.* BASF. Purdue University.

2025 **Cong, Yan**. *Humans in Charge, AI in the Loop.* Cornerstone retreat. College of Liberal Arts. Purdue University.

2025 **Cong, Yan**. *Reasoning with Enriched Meaning.* SLC colloquium. School of Languages and Cultures. Purdue University.

2024 **Cong, Yan**; Issa, Elsayed; Rayz, Julia T. *Leveraging AI in Language Learning.* College of Liberal Arts Research Academy (CLARA) Research Day. Purdue University.

2023 **Cong, Yan**. *Exploring L2 Writing Proficiency via LLMs Metrics: a Preliminary Study. LINGTalk.* Department of Linguistics, Purdue University.

2023 **Cong, Yan**. *Using large language models to measure fluency.* Advanced Methods at Purdue (AMAP) in the behavioral, health, and social sciences. Purdue University.

2023 **Cong, Yan**; Nikzad, Amir; Tang, Sunny X. *Multi-modal computational speech and language features for assessing psychiatric conditions.* The University of Maryland.

2023 **Cong, Yan**; Hänsel, Katrin; Nikzad, Amir; Tang, Sunny X. *Investigating novel NLP approaches for measuring speech disturbances in psychosis.* The Functional Neuroimaging and Bioinformatics Lab. McLean, Harvard Medical School Affiliate.

2021 **Cong, Yan**. *Formal Linguistics and Philosophy (FLaP).* What we don't say matters. The Hong Kong Baptist University, Department of English.

Fellowships, Funds, and Awards

2025 **John Martinson Honors College, Research Breakthrough Award**, *Developing self-managed story-based treatment for aphasia and AI-assisted predictors for language recovery*, Purdue University. PI: Jiyeon Lee. Co-PI: Catherine Pham and Yan Cong. $60,000

2024 **Systems Fellows Award**, Purdue Systems Collaboratory, Purdue University.

2024 **Internal fund**, Faculty Research Development Award.
Leveraging large language models for aphasia detection and prediction in older adults. Center on Aging and the Life Course (CALC), Purdue University. $10,000

2023 **Educational Advancement Award**
Computational approach to speech biomarkers in bilingual immigrants with psychosis.
Advancing Women in Science and Medicine, Northwell health. $5,000

2022 **Educational Advancement Award**
Computational linguistics can predict psychiatric diagnoses and outcome.
Advancing Women in Science and Medicine, Northwell health. $5,000

2020 **Dissertation Completion Fellowship**
MSU Graduate School $10,000

2018 **Student Fellowship**, *North American Summer School on Logic, Language, and Information (NASSLLI)*, at Pittsburgh, Pennsylvania
Carnegie Mellon University $500

2017-2020 **Summer Support Fellowship**
MSU Department of Linguistics and Languages $4,500

2015 **Best Paper Award**, *The 29th Pacific Asia Conference on Language, Information and Computation*, The invertible construction in Chinese. Shanghai Jiaotong University, China

Teaching

Undergraduate
- LC490-LING398: Language Learning and AI, Purdue (2024-present)
- LC490: Language research and AI (Independent study), Purdue (2025-present)
- SCLA102: Cornerstone - Critical thinking and communication, Purdue (2023-present)
- LIN401: Introduction to Linguistics, MSU (2017-2021)
- CHS101, CHS102, CHS411: Chinese language, MSU (2016-2017)

Graduate
- LING598-LC596: Statistical Modeling in Linguistics II, Purdue (2025-present)
- LING598-LC596: Statistical Modeling in Linguistics, Purdue (2023-present)

Advising

Graduate student committee member
- SLC / Linguistics: Yue Li; Jingying Hu; Daeun Kang; Karen Fullin.
- Computer and Information Technology: Yifei Hu; Damin Zhang.
- Civil Engineering: Sukrati Gautam (Purdue Systems Collaboratory Fellow advisor)
- Speech and Language Pathology: Peng Zhang; Alton Chao.

Undergraduate Involvement in research

- Emily A. Tumacder: Presented at the Spring 2025 Undergraduate Research Conference: Engineering, April 08.
- Cameron Pilla: Machine learning, large language models deployment
- Lauren Kless: AI tools and methods for understanding language decline and recovery (Spring 2025; Independent study LC490)
- Trisha Godara: EACL 2026 Student Research Workshop (SRW) Grant; NLP pipelines development and optimization, vision and language models

Service to the Profession

2025-present Guest Editor: Large language models in psychological science. *Nature Humanities and Social Sciences Communications.*

2024-present Editorial Board Member: *Humanities and Social Sciences Communications, Springier Nature portfolio* (HSS).

2024-present Program Committee and Reviewer:

- *The Workshop on Innovative Use of NLP for Building Educational Applications* (BEA 2025, 2026).
- *The 3rd edition of the BriGap workshop: Bridges and Gaps between Formal and Computational Linguistics* (BriGap 2026).
- *The 2026 International Conference on Language Resources and Evaluation* (LREC 2026).
- *The 39th Pacific Asia Conference on Language, Information and Computation* (PACLIC 2025).
- *The 31st International Conference on Computational Linguistics* (COLING 2024).
- *COLM 2025 Workshop Pragmatic Reasoning in Language Models* (PragLM).

2022-present Reviewer (Ad hoc):

- *The Association for Computational Linguistics* (ACL Rolling Review)
- *Language learning*
- *Research Methods in Applied Linguistics*
- *Nature Communications*
- *Nature Portfolio*
- *Nature Scientific Reports*
- *Nature Communications Medicine*
- *Advanced science*
- *PLOS One*
- *Association of Contemporary African Linguistics* (ACAL)
- *American Journal of Speech-Language Pathology (AJSLP)*
- *npj Mental Health*
- *Journal of Speech, Language, and Hearing Research (JSLHR)*
- *Health Informatics Journal*
- *Journal of Medical Internet Research*
- *Purdue Linguistics Symposium Proceedings*

Service to Purdue University

Member	Graduate admissions subcommittee.	since Fall 2024
Member	CLA Ross-Lynn Postdoctoral Fellow review subcommittee.	for Fall 2024
Organizer	Computation and Linguistic Meaning Lab.	since Fall 2023

Non-Degree Education

2020 North American Summer School for Logic, Language, and Information (NASSLLI), Brandeis University (Online due to COVID-19)

2019 Linguistics Society of America (LSA) Summer Institute, University of California, Davis

2018 Crete Summer School in Linguistics, University of Crete

2018 NASSLLI, Carnegie Mellon University

2017 LSA Summer Institute, University of Kentucky

March 7, 2026